Visualization Literacy Analysis

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Read in data files

```
#grab files from google drive (only have to do this once)
# source("getFromGoogleDrive.R")

#get the first 6? characters of each data file
#get unique values of these
# this is list of subject ids

raw_file_names <- list.files("../AccData")
first_six <- substr(raw_file_names, 1, 6)
sub_ids <- unique(first_six)


# fast_RTs <- data.frame(ParticipantId = character(),
#                         TrialName = character(),
#                         type = character(),
#                         time = numeric()
# )

all_data <- NULL

for (i in 1:length(sub_ids)){

  if (grepl("~$", sub_ids[i], fixed = T)){
    next
  }

  temp_file1 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_1.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  temp_file2 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_2.xlsx")) %>%
    slice(1:17) %>%
     select(-starts_with("Order")) %>%
    rename(correct = 8)

  new_temp <- temp_file1 %>%
    bind_cols(temp_file2$correct) %>%
    rename(Correct_1 = correct, Correct_2 = "...9") %>%
    mutate(AnswerRT = TimeToBeginInput - TimeToReadQuestion)
```

```
  all_data <- all_data %>%
    bind_rows(new_temp)

}

all_data <- all_data %>%
  rename(readRT = TimeToReadQuestion, totalRT = TimeToBeginInput)

#read in trialtype key (I created this from an early version of the previous paper)
trial_type_key <- read.csv("../trial_type_key.csv", stringsAsFactors = F)

all_data <- all_data %>%
  mutate(TrialType = trial_type_key$TrialType[match(TrialName, trial_type_key$TrialName)]) %>%
  mutate(TrialType = paste0("Type",TrialType))
```

Basic checks

```
#how many participants per condition
all_data %>%
  group_by(ParticipantId, Condition) %>%
  summarize(ntrials = n()) %>%
  group_by(Condition) %>%
  summarize(nsubs = n()) %>%
  kable()
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

Condition	nsubs
VR	50
VR Monitor	39
VR Monitor Stereo	33

```
#why is the balance so off?
```

Remove outliers based on RT

```
#removing on trial-by-trial basis


#remove answerRTs below 2000ms first
all_data_remove <- all_data %>%
  filter(AnswerRT >= 2000)
dim(all_data)[1]
```

```
## [1] 2074
```

```
dim(all_data_remove)[1]
```

```
## [1] 2067
```

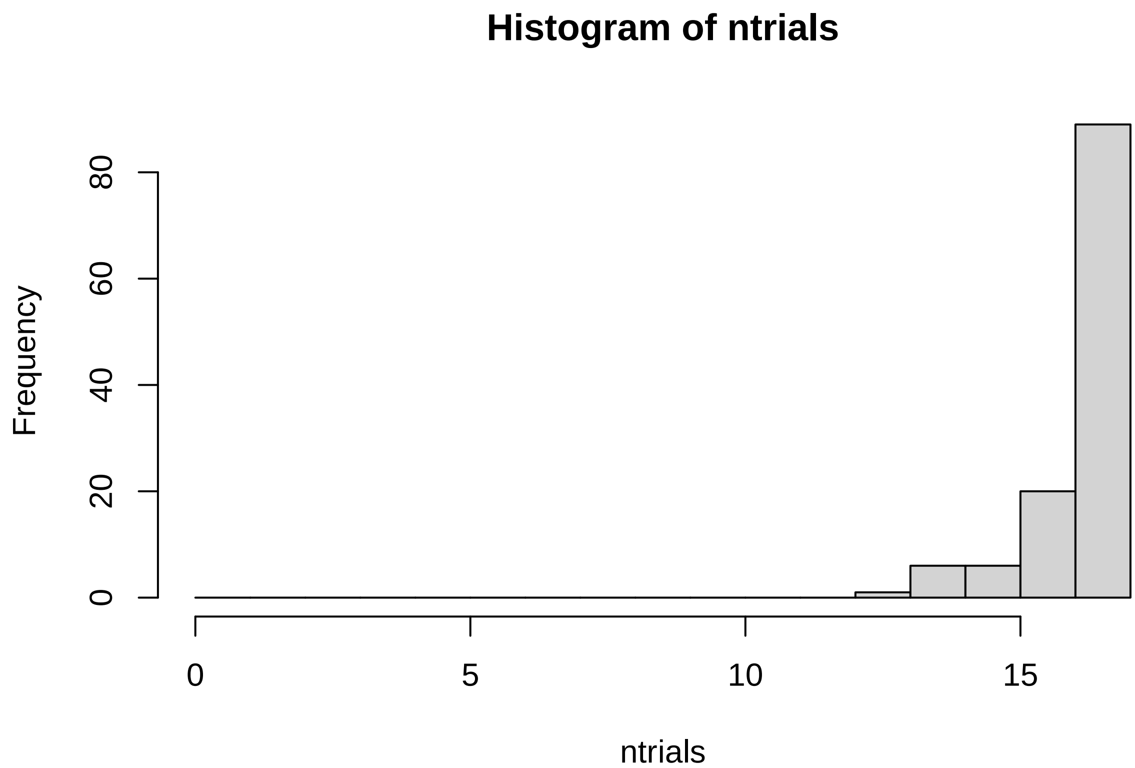
```r
#drops 7 trials

rt_data_summary <- all_data %>%
  group_by(TrialName) %>%
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),
            sdAnswerRT = sd(AnswerRT, na.rm = T),
            UB = meanAnswerRT + 3*sdAnswerRT,
            LB = meanAnswerRT - 3*sdAnswerRT)
rt_data_summary
```

```
## # A tibble: 17 x 5
##    TrialName     meanAnswerRT sdAnswerRT      UB      LB
##    <chr>                <dbl>      <dbl>   <dbl>   <dbl>
##  1 BarChartQ1          27611.     14762.  71897. -16675.
##  2 BarChartQ2          16921.     13338.  56935. -23093.
##  3 BarChartQ3          15609.      9948.  45452. -14235.
##  4 BarChartQ4          10288.      8978.  37222. -16646.
##  5 LineChartQ1         49185.     29505. 137699. -39329.
##  6 LineChartQ2         40127.     31660. 135107. -54854.
##  7 LineChartQ3         27190.     17504.  79701. -25322.
##  8 LineChartQ4         14779.     16568.  64483. -34925.
##  9 LineChartQ5         27877.     17250.  79628. -23874.
## 10 ScatterplotQ1       32801.     24294. 105682. -40080.
## 11 ScatterplotQ2       29636.     23980. 101576. -42304.
## 12 ScatterplotQ3       45580.     33014. 144623. -53463.
## 13 ScatterplotQ4       35987.     25402. 112194. -40219.
## 14 ScatterplotQ5       59805.     42684. 187859. -68248.
## 15 SurfacePlotQ1       56608.     36042. 164733. -51516.
## 16 SurfacePlotQ2       65110.     50062. 215297. -85076.
## 17 SurfacePlotQ3       48373.     37424. 160646. -63900.
```

```r
all_data_no_outliers <- all_data_remove %>%
  group_by(TrialName) %>%
  filter((!(abs(AnswerRT - mean(AnswerRT)) > 3.0*sd(AnswerRT))))
dim(all_data_no_outliers)[1]
```

```
## [1] 2020
```

```r
#drops 47 more trials

all_data_no_outliers %>%
  group_by(ParticipantId) %>%
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:17))
```

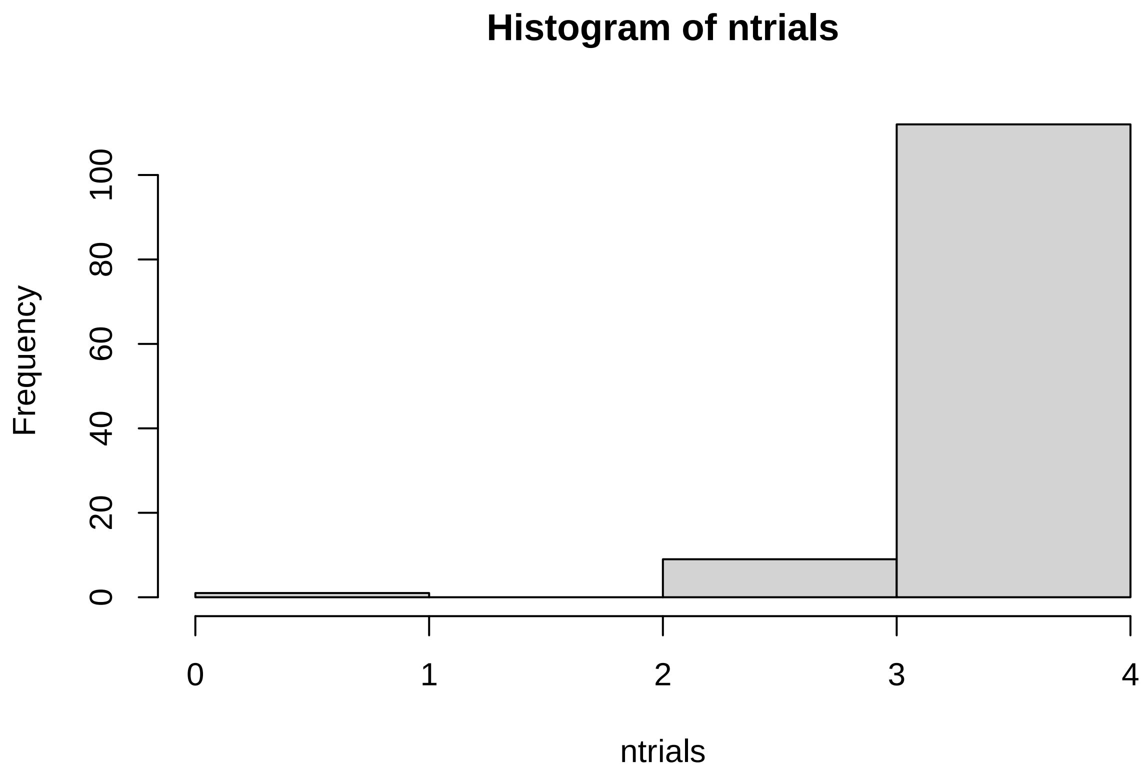



```
##maybe consider replacing outliers with means instead of removing them?
```

Extract Only One Type of Chart

Create a dataframe from the data with no outliers that contains only XChartQ1 through XChartQ4 trials for the following analysis.

```
all_data_no_outliers <- all_data_no_outliers %>%
  filter(grepl("BarChartQ[0-9]", TrialName))

all_data_no_outliers %>%
  group_by(ParticipantId) %>%
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:4))
```



Compare RTs for conditions and question type (three types: identify, relate, predict)

```
#compare read times (should be no differences of condition)
#compare answer times (potentially a difference)


trial_type_means <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_readRT = mean(readRT),
            mean_answerRT = mean(AnswerRT),
            n = n())
```

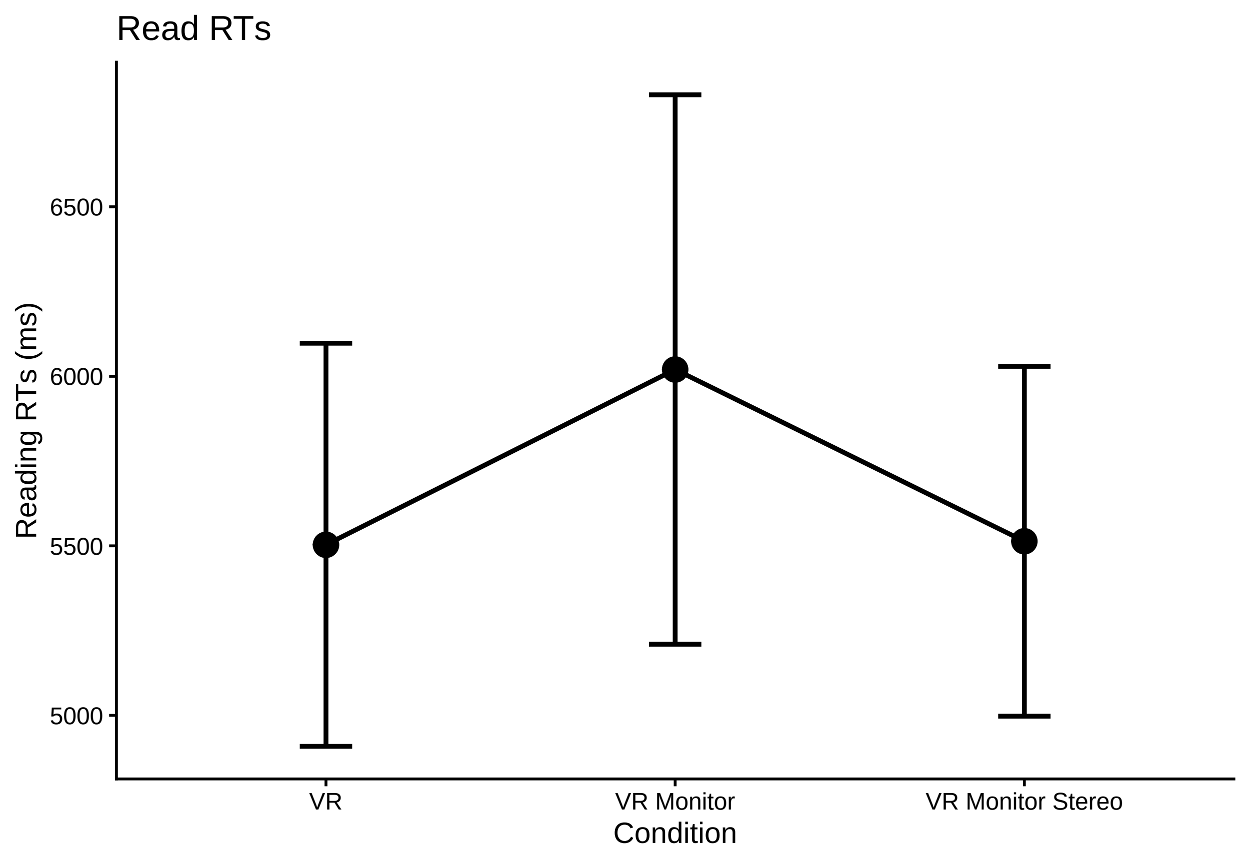
```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
## the `.groups` argument.
```

```
# first, make .csv files in wide format to double check in statview
read_rt_type_wider <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_readRT) %>%
  pivot_wider(names_from = TrialType,values_from = mean_readRT)

# fill in na values with mean for type1, type2 and type3 RT
# the following is supposed to work, but we get NaN from calculation of mean here, not sure why...
#read_rt_type_wider_clean <- read_rt_type_wider %>% mutate(across(Type1, ~replace_na(., mean(., na.rm=T
read_rt_type_wider_clean <- read_rt_type_wider %>% replace_na(list(Type1 = mean(read_rt_type_wider$Type
read_rt_type_wider_clean <- read_rt_type_wider_clean %>% replace_na(list(Type2 = mean(read_rt_type_wide
```

```r
read_rt_type_wider_clean <- read_rt_type_wider_clean %>% replace_na(list(Type3 = mean(read_rt_type_wide

answer_rt_type_wider <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_answerRT) %>%
  pivot_wider(names_from = TrialType,values_from = mean_answerRT)
write.csv(read_rt_type_wider, file = "readtypeRTs.csv", row.names = F)
write.csv(answer_rt_type_wider, file = "answertypeRTs.csv", row.names = F)

# fill in na values with mean for type1, type2 and type3 RT
answer_rt_type_wider_clean <- answer_rt_type_wider %>% replace_na(list(Type1 = mean(answer_rt_type_wide
answer_rt_type_wider_clean <- answer_rt_type_wider_clean %>% replace_na(list(Type2 = mean(answer_rt_typ
answer_rt_type_wider_clean <- answer_rt_type_wider_clean %>% replace_na(list(Type3 = mean(answer_rt_typ


#### READ RTs ####

#make some plots

#just condition main effect
readplot1 <- all_data_no_outliers %>%
   group_by(ParticipantId, Condition) %>%
   summarize(overallmean = mean(readRT)) %>%
   group_by(Condition) %>%
   summarize(overall_condition_mean = mean(overallmean),
             se = std.error(overallmean),
             n = n(),
             CI = qt(0.975,df=n-1)*se)
```

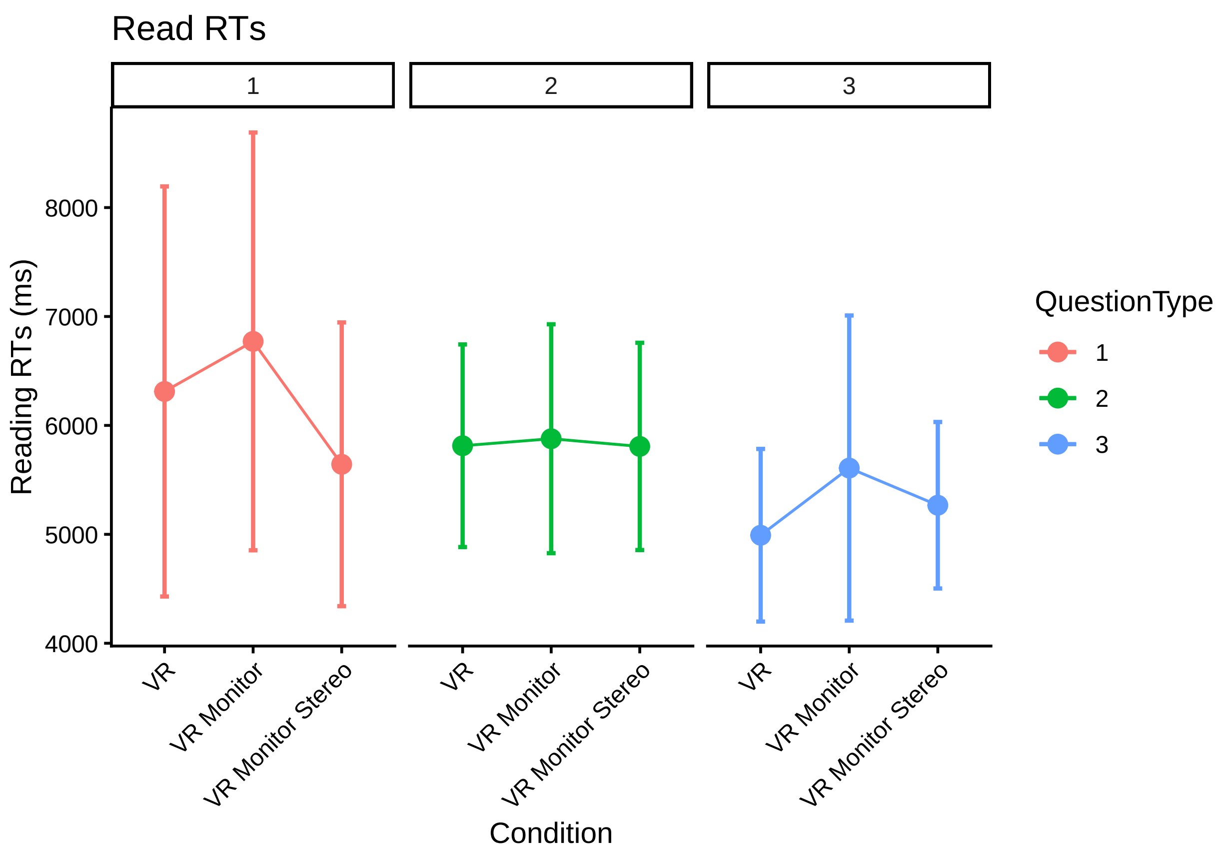
```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

```r
 ggplot(readplot1, aes(Condition,
                      overall_condition_mean,
                      group = 1,
                      ymin = overall_condition_mean - CI,
                      ymax = overall_condition_mean + CI)) +
   theme_classic() +
   geom_point(size = 4) +
   geom_errorbar(width = .15, linewidth = 0.85) +
   geom_line(linewidth = 0.85) +
   labs(y = "Reading RTs (ms)", title = "Read RTs")
```




```r
#make a little plot using wide format
superbPlot(read_rt_type_wider_clean,
    BSFactors    = "Condition",
    WSFactors    = "QuestionType(3)",
    variables    = c("Type1", "Type2", "Type3"),
    statistic    = "meanNArm",
    errorbar     = "CI",
    gamma        = 0.95,
    adjustments = list(
        purpose       = "difference"
    ),
    plotLayout = "line",
    factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Reading RTs (ms)", title = "Read RTs")
```

```
## superb::ADVICE: Some of the groups' variances are heterogeneous. Consider using purpose="tryon".
```




```
# there are missing values in anova for some participants because they don't
# do all trial types necessarily.  Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_means %>%
     select(ParticipantId, Condition, TrialType, mean_readRT) %>%
     pivot_wider(names_from = TrialType, values_from = mean_readRT)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
readRT_means_clean <- x_clean %>%
     pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_readRT")

read_rt_type_anova <- aov_ez(id = "ParticipantId",
                             dv = "mean_readRT",
                             data = readRT_means_clean,
                              within = "TrialType",
                              between = "Condition",
                             anova_table = list(es = "pes") #might want to double-check these
)
```

```
## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
```

```
kable(nice(read_rt_type_anova))
```

Effect	df	MSE	F	pes	p.value
Condition	2, 119	14643620.63	0.55	.009	.581
TrialType	1.58, 188.04	7498202.73	4.57 *	.037	.018
Condition:TrialType	3.16, 188.04	7498202.73	0.65	.011	.594

```
#posthoc test for trial type
pairs(emmeans(read_rt_type_anova, "TrialType"), adjust = "Tukey")
```

```
##  contrast      estimate  SE  df t.ratio p.value
##  Type1 - Type2      409 330 119   1.238  0.4333
##  Type1 - Type3      953 373 119   2.553  0.0318
##  Type2 - Type3      544 227 119   2.397  0.0472
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#posthoc test for condition, still using Tukey
pairs(emmeans(read_rt_type_anova, "Condition"), adjust = "Tukey")
```

```
##  contrast                         estimate  SE  df t.ratio p.value
##  VR - VR Monitor                      -380 472 119  -0.806  0.7002
##  VR - VR Monitor Stereo                133 496 119   0.269  0.9610
##  VR Monitor - VR Monitor Stereo        513 523 119   0.982  0.5894
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates
```

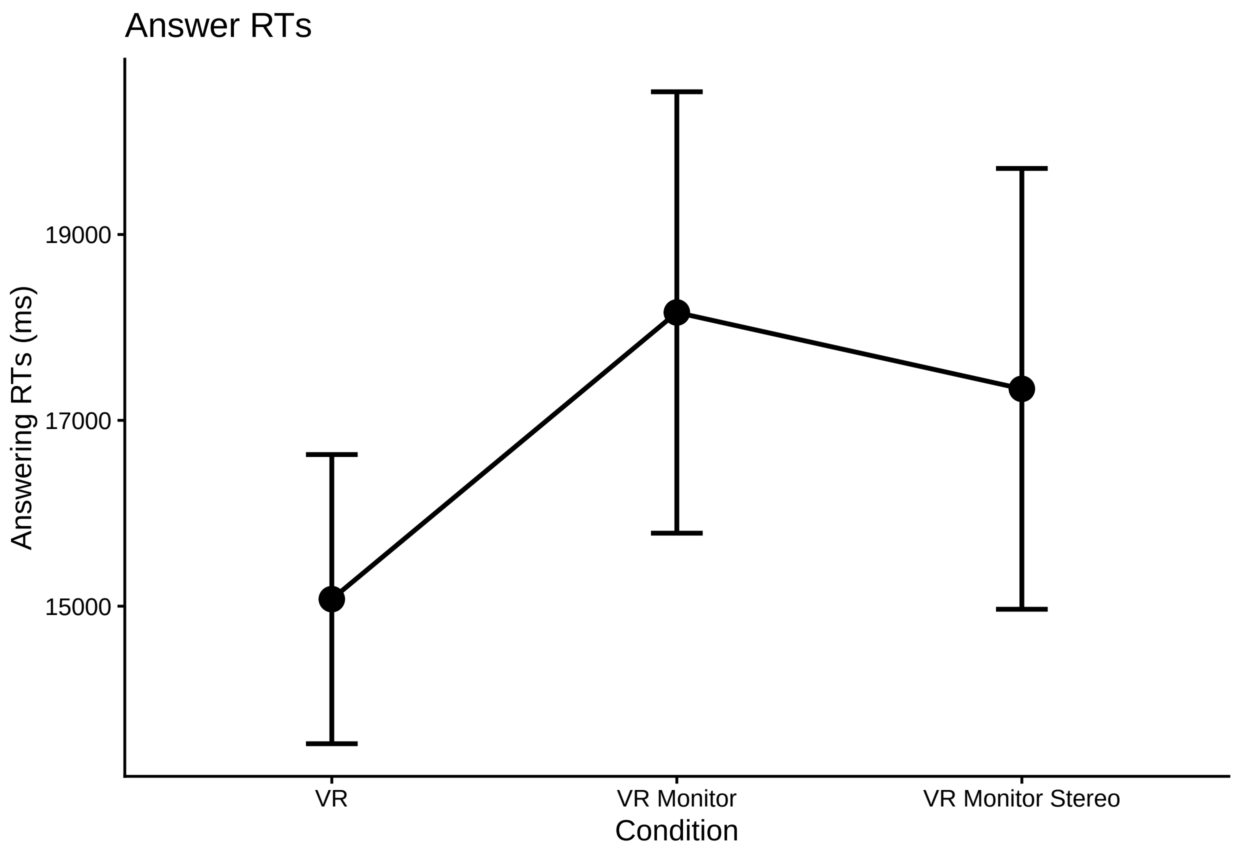
```
#posthoc test for condition, others seem to be scheffe, sidak, bonferroni, dunnettx, mvt, none
pairs(emmeans(read_rt_type_anova, "Condition"), adjust = "scheffe")
```

```
##  contrast                         estimate  SE  df t.ratio p.value
##  VR - VR Monitor                      -380 472 119  -0.806  0.7234
##  VR - VR Monitor Stereo                133 496 119   0.269  0.9646
##  VR Monitor - VR Monitor Stereo        513 523 119   0.982  0.6184
##
## Results are averaged over the levels of: TrialType
## P value adjustment: scheffe method with rank 2
```

```
#### ANSWER RTs ####

#make some plots

#just condition main effect
answerplot1 <- all_data_no_outliers %>%
   group_by(ParticipantId, Condition) %>%
   summarize(overallmean = mean(AnswerRT)) %>%
   group_by(Condition) %>%
   summarize(overall_condition_mean = mean(overallmean),
             se = std.error(overallmean),
             n = n(),
             CI = qt(0.975,df=n-1)*se)
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

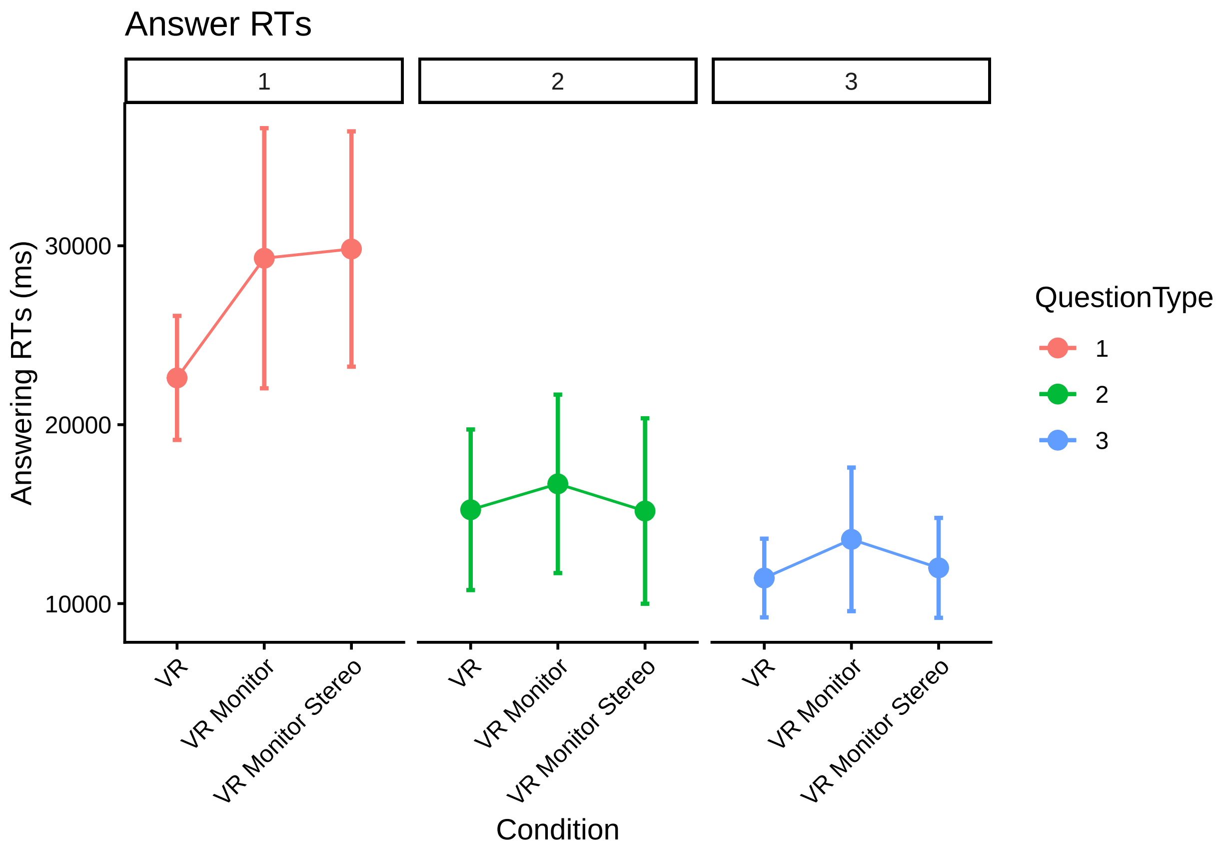
```r
ggplot(answerplot1, aes(Condition,
                       overall_condition_mean,
                       group = 1,
                       ymin = overall_condition_mean - CI,
                       ymax = overall_condition_mean + CI)) +
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Answering RTs (ms)", title = "Answer RTs")
```




```r
#make the little plot
superbPlot(answer_rt_type_wider_clean,
    BSFactors    = "Condition",
    WSFactors    = "QuestionType(3)",
    variables    = c("Type1", "Type2", "Type3"),
    statistic    = "meanNArm",
    errorbar     = "CI",
    gamma        = 0.95,
    adjustments = list(
        purpose        = "difference"
    ),
    plotLayout = "line",
    factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
```

```r
theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Answering RTs (ms)", title = "Answer RTs")
```

```
## superb::ADVICE: Some of the groups' variances are heterogeneous. Consider using purpose="tryon".
```




```r
# there are missing values in anova for some participants because they don't
# do all trial types necessarily.  Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_means %>%
     select(ParticipantId, Condition, TrialType, mean_answerRT) %>%
     pivot_wider(names_from = TrialType, values_from = mean_answerRT)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
answerRT_means_clean <- x_clean %>%
     pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_answerRT")

answer_rt_type_anova <- aov_ez(id = "ParticipantId",
                               dv = "mean_answerRT",
                               data = answerRT_means_clean,
                                within = "TrialType",
                                between = "Condition",
                               anova_table = list(es = "pes") #might want to double-check these
```

```
)
```

```
## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
```

```r
kable(nice(answer_rt_type_anova))
```

Effect	df	MSE	F	pes	p.value
Condition	2, 119	148946021.28	2.87 +	.046	.060
TrialType	1.76, 209.64	97205116.20	84.60 ***	.416	<.001
Condition:TrialType	3.52, 209.64	97205116.20	2.12 +	.034	.088

```r
#posthoc test for trial type
pairs(emmeans(answer_rt_type_anova, "TrialType"), adjust = "Tukey")
```

```
##  contrast      estimate   SE  df t.ratio p.value
##  Type1 - Type2    11542 1370 119   8.424  <.0001
##  Type1 - Type3    14906 1220 119  12.184  <.0001
##  Type2 - Type3     3364  980 119   3.432  0.0024
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#Question Type 3 much faster than Type 1/Type 2 (this is answering times)

#posthoc test for condition, still using Tukey
pairs(emmeans(answer_rt_type_anova, "Condition"), adjust = "Tukey")
```

```
##  contrast                        estimate   SE  df t.ratio p.value
##  VR - VR Monitor                    -3433 1510 119  -2.281  0.0625
##  VR - VR Monitor Stereo             -2568 1580 119  -1.625  0.2391
##  VR Monitor - VR Monitor Stereo       865 1670 119   0.519  0.8622
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
ref <- emmeans(answer_rt_type_anova,~Condition|TrialType)

pairs(ref, adjust = "Tukey")
```

```
## TrialType = Type1:
##  contrast                        estimate   SE  df t.ratio p.value
##  VR - VR Monitor                  -6689.0 2680 119  -2.497  0.0367
##  VR - VR Monitor Stereo           -7205.0 2810 119  -2.562  0.0311
##  VR Monitor - VR Monitor Stereo   -516.0 2970 119  -0.174  0.9835
##
## TrialType = Type2:
##  contrast                        estimate   SE  df t.ratio p.value
##  VR - VR Monitor                  -1448.8 2320 119  -0.625  0.8069
##  VR - VR Monitor Stereo              69.6 2440 119   0.029  0.9996
##  VR Monitor - VR Monitor Stereo   1518.4 2570 119   0.591  0.8251
##
## TrialType = Type3:
##  contrast                        estimate   SE  df t.ratio p.value
##  VR - VR Monitor                  -2161.3 1440 119  -1.506  0.2918
```

```
##   VR - VR Monitor Stereo           -568.6 1510 119  -0.377  0.9245
##   VR Monitor - VR Monitor Stereo   1592.7 1590 119   1.002  0.5768
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#plot and interaction suggests that the conditions have different effects for different
#question types. posthoc tests indicate a difference between VR and VRMonitor for QType 1
```

Accuracy

```
# do interrater reliability: Fits each trial as independent with two raters
# That is, this ignores participants (complete pooling, which may be problematic)
# 1410 trials from N = 85 participants, not exactly 17 trials per participant
# b/c of excluded trials
  irr::icc(select(ungroup(all_data_no_outliers), Correct_1, Correct_2))
```

```
##  Single Score Intraclass Correlation
##
##    Model: oneway
##    Type : consistency
##
##    Subjects = 476
##      Raters = 2
##      ICC(1) = 0.91
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(475,476) = 21.3 , p = 1.58e-184
##
##  95%-Confidence Interval for ICC Population Values:
##   0.893 < ICC < 0.924
```

```
  head(all_data_no_outliers)
```

```
## # A tibble: 6 x 11
## # Groups:   TrialName [4]
##   ParticipantId Condition  TrialNumber TrialName  readRT totalRT UserInput       Correct_1
##   <chr>         <chr>            <dbl> <chr>       <dbl>   <dbl> <chr>               <dbl>
## 1 AA0176        VR Monitor          11 BarChartQ1  4669.  21882. North in 2010           1
## 2 AA0176        VR Monitor          12 BarChartQ2  7003.  15783. Remians stable~         1
## 3 AA0176        VR Monitor          13 BarChartQ3  6200.  10132. North neighbor~         1
## 4 AA0176        VR Monitor          14 BarChartQ4  5015.  11504. West neighborh~         1
## 5 AA0876        VR                   9 BarChartQ1  3519.  32929. 2010 north              1
## 6 AA0876        VR                  10 BarChartQ2  4759.  23479. stable                  1
## # i 3 more variables: Correct_2 <dbl>, AnswerRT <dbl>, TrialType <chr>
```

```
# account for non-independence?
  #Data has to be narrow
#https://cran.r-project.org/web/packages/cccrm/index.html
icc.dat.narrow <- all_data_no_outliers %>%
                    pivot_longer(cols = c('Correct_1', 'Correct_2'))

colnames(icc.dat.narrow)[10] <- "Rater"
icc.dat.narrow$Rater <- as.factor(icc.dat.narrow$Rater)

#Longitudinal concordance for two raters
```

```r
#By all 17 trials (by items)
# icc.rm.trial.name <- ccclon(icc.dat.narrow,
#                              "value", "ParticipantId", "TrialName", "Rater")
#  icc.rm.trial.name
#  summary(icc.rm.trial.name)

#By time using trial numbers (by time)
 icc.rm.trial.number <- ccclon(icc.dat.narrow,
                               "value", "ParticipantId", "TrialNumber", "Rater" )
```

```
## Warning: `ccclon()` was deprecated in cccrm 3.0.0.
## i Please use `ccc_vc()` instead.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.
```

```r
 icc.rm.trial.number
```

```
## CCC estimated by variance components:
##         CCC    LL CI 95%   UL CI 95%      SE CCC
## 0.913965024 0.895570977 0.929240100 0.008544854
```

```r
 summary(icc.rm.trial.number)
```

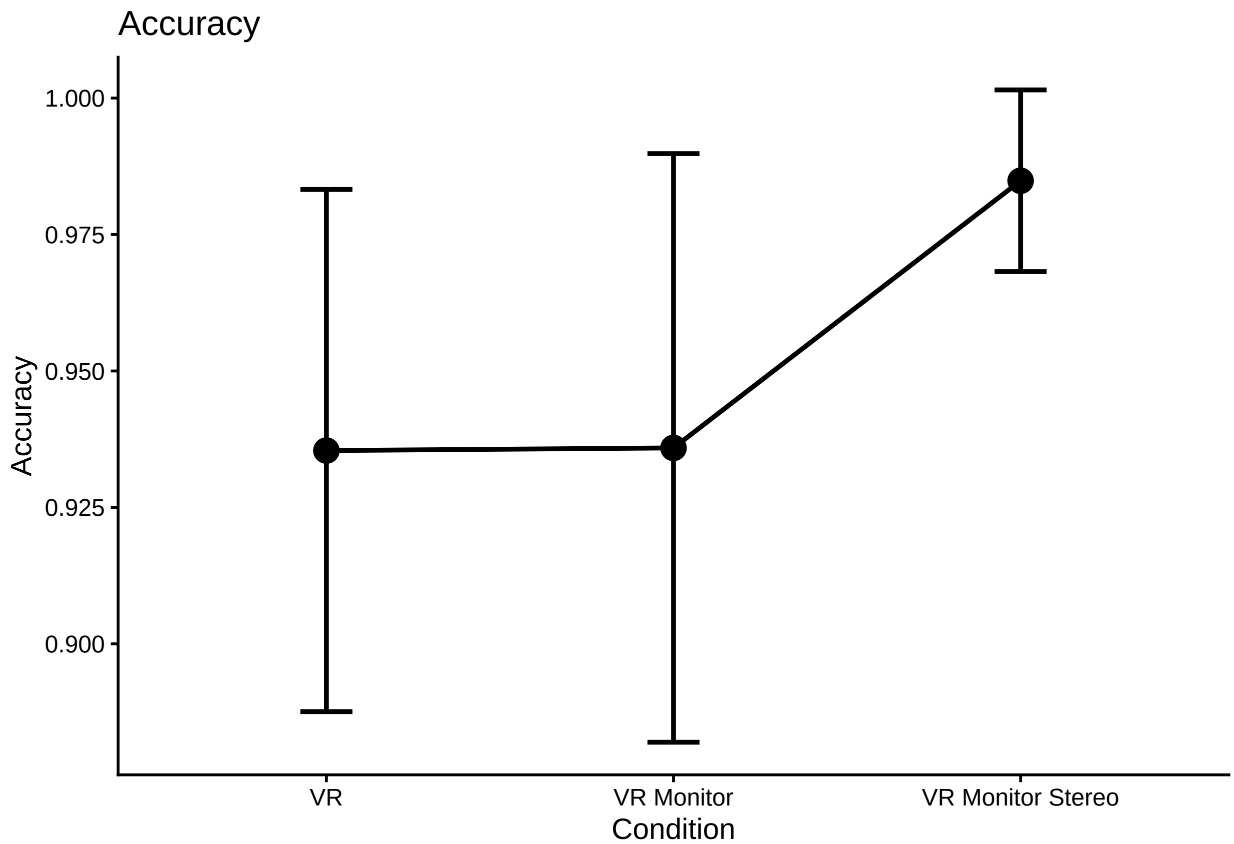
```
##        Subjects Subjects-Method   Subjects-Time          Method           Error
##    1.348428e-02    7.407619e-10    2.525883e-02    0.000000e+00    3.647035e-03
##
## CCC estimated by variance components
##         CCC    LL CI 95%   UL CI 95%      SE CCC
## 0.913965024 0.895570977 0.929240100 0.008544854
```

```r
# A similar approach
#https://peerj.com/articles/9850/
# test.mod <-
#  lcc(data = hue, subject = "Fruit", resp = "H_mean",
#                  method = "Method", time = "Time", qf = 2, qr = 1)
# summary(test.mod)

# icc.dat.narrow$ParticipantId <- as.factor(icc.dat.narrow$ParticipantId)
# icc.dat.narrow$Rater <- as.factor(icc.dat.narrow$Rater)
#
#
# icc.rm.another <-
# lcc(data = icc.dat.narrow,
#       subject = "ParticipantId",
#       resp = "value",
#       method = "Rater",
#       time = "TrialNumber",
#       qf = 1, #polynomial trends (1 to 3)
#       qr = 0, #random effects (0 is random int only, default)
#       components = TRUE)
#
# #LCC = Longitudinal Concordance Correlation
# #LPC = Longitudinal Pearson Correlation
# #LA  = Longitudinal Accuracy
# icc.rm.another
#
# #Model coefficients near 1 but only the the last one has a correctly
```

```r
# #estimated model. Maybe because it's more or less flat over TrialNumber?
# lccPlot(icc.rm.another, type = "lpc")
# lccPlot(icc.rm.another, type = "lcc")
# lccPlot(icc.rm.another, type = "la")


all_data_no_outliers <- all_data_no_outliers %>%
  rowwise() %>%
  mutate(Correct_Avg = mean(c(Correct_1, Correct_2)))

#just condition main effect
accplot1 <- all_data_no_outliers %>%
   group_by(ParticipantId, Condition) %>%
   summarize(overallmean = mean(Correct_Avg)) %>%
   group_by(Condition) %>%
   summarize(overall_condition_mean = mean(overallmean),
             se = std.error(overallmean),
             n = n(),
             CI = qt(0.975,df=n-1)*se)
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

```r
 ggplot(accplot1, aes(Condition,
                       overall_condition_mean,
                       group = 1,
                       ymin = overall_condition_mean - CI,
                       ymax = overall_condition_mean + CI)) +
   theme_classic() +
   geom_point(size = 4) +
   geom_errorbar(width = .15, linewidth = 0.85) +
   geom_line(linewidth = 0.85) +
   labs(y = "Accuracy", title = "Accuracy")
```



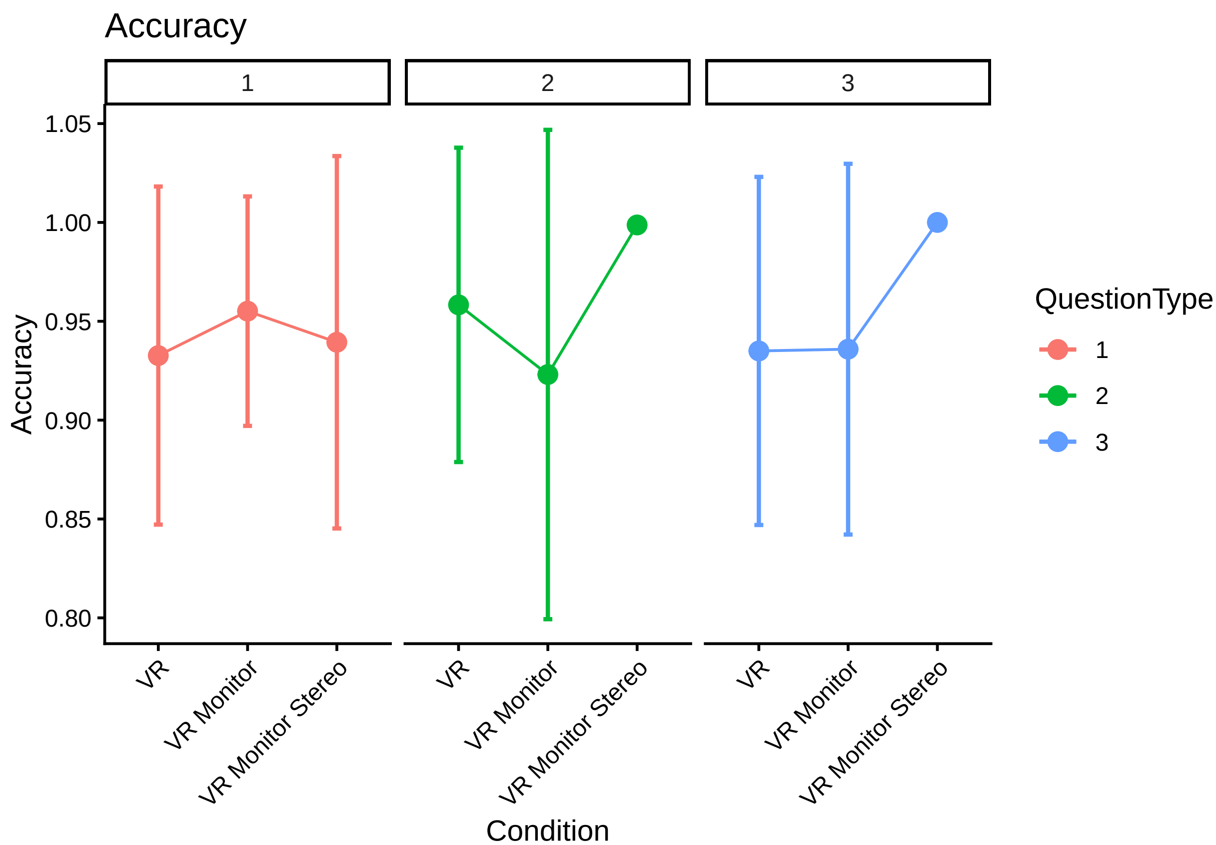

```
trial_type_mean_acc <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_acc = mean(Correct_Avg),
            n = n())
```

```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
## the `.groups` argument.
```

```
acc_type_wider <- trial_type_mean_acc %>%
  select(ParticipantId, Condition, TrialType, mean_acc) %>%
  pivot_wider(names_from = TrialType,values_from = mean_acc)
write.csv(acc_type_wider, file = "typeacc.csv", row.names = F)

# replace na with mean
acc_type_wider_clean <- acc_type_wider %>% replace_na(list(Type1 = mean(acc_type_wider$Type1, na.rm=TRUI
acc_type_wider_clean <- acc_type_wider_clean %>% replace_na(list(Type2 = mean(acc_type_wider$Type2, na.:
acc_type_wider_clean <- acc_type_wider_clean %>% replace_na(list(Type3 = mean(acc_type_wider$Type3, na.:

#make the little plot
superbPlot(acc_type_wider_clean,
    BSFactors    = "Condition",
    WSFactors    = "QuestionType(3)",
    variables    = c("Type1", "Type2", "Type3"),
    statistic    = "meanNArm",
    errorbar     = "CI",
    gamma        = 0.95,
    adjustments = list(
```

```r
        purpose       = "difference"
    ),
    plotLayout = "line",
    factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Accuracy", title = "Accuracy")
```

```
## superb::ADVICE: Some of the groups' variances are heterogeneous. Consider using purpose="tryon".
```




```r
# there are missing values in anova for some participants because they don't
# do all trial types necessarily.  Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_mean_acc %>%
     select(ParticipantId, Condition, TrialType, mean_acc) %>%
     pivot_wider(names_from = TrialType, values_from = mean_acc)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
acc_means_clean <- x_clean %>%
     pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_acc")
```

```r
acc_type_anova <- aov_ez(id = "ParticipantId",
                         dv = "mean_acc",
                         data = acc_means_clean,
                          within = "TrialType",
                          between = "Condition",
                         anova_table = list(es = "pes") #might want to double-check these
)
```

```
## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
```

```r
kable(nice(acc_type_anova))
```

Effect	df	MSE	F	pes	p.value
Condition	2, 119	0.06	0.98	.016	.378
TrialType	1.98, 235.25	0.02	0.43	.004	.648
Condition:TrialType	3.95, 235.25	0.02	1.05	.017	.382

```r
#posthoc test for trial type
pairs(emmeans(acc_type_anova, "TrialType"), adjust = "Tukey")
```

```
##  contrast      estimate     SE  df t.ratio p.value
##  Type1 - Type2 -0.01764 0.0192 119  -0.919  0.6291
##  Type1 - Type3 -0.01457 0.0207 119  -0.703  0.7622
##  Type2 - Type3  0.00308 0.0210 119   0.147  0.9882
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates
```

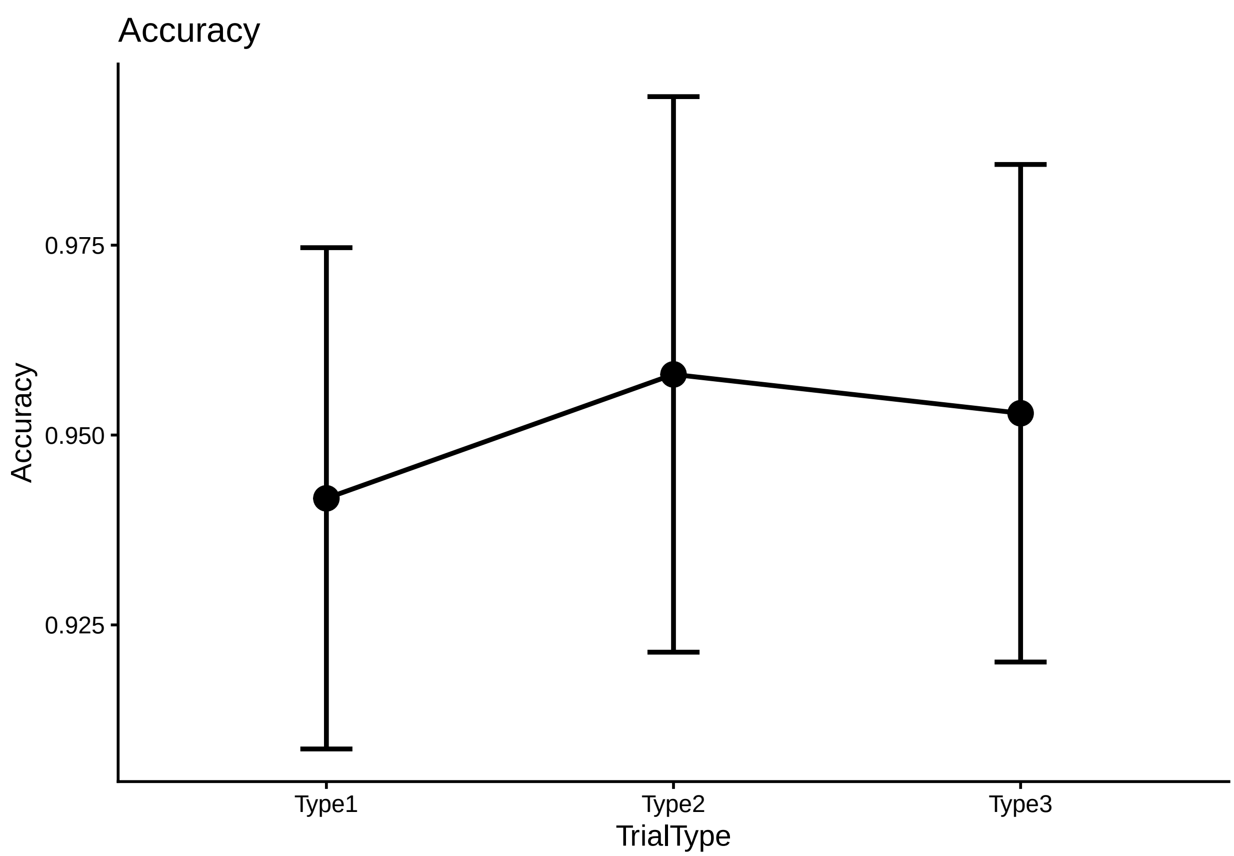
```r
#Question Type 3 much faster than Type 1/Type 2 (this is answering times)

ref <- emmeans(acc_type_anova,~Condition|TrialType)

pairs(ref, adjust = "Tukey")
```

```
## TrialType = Type1:
##  contrast                        estimate     SE  df t.ratio p.value
##  VR - VR Monitor                -0.022462 0.0389 119  -0.577  0.8328
##  VR - VR Monitor Stereo         -0.006727 0.0409 119  -0.165  0.9852
##  VR Monitor - VR Monitor Stereo  0.015734 0.0431 119   0.365  0.9293
##
## TrialType = Type2:
##  contrast                        estimate     SE  df t.ratio p.value
##  VR - VR Monitor                 0.035242 0.0424 119   0.831  0.6845
##  VR - VR Monitor Stereo         -0.040407 0.0445 119  -0.908  0.6365
##  VR Monitor - VR Monitor Stereo -0.075650 0.0469 119  -1.611  0.2448
##
## TrialType = Type3:
##  contrast                        estimate     SE  df t.ratio p.value
##  VR - VR Monitor                -0.000897 0.0389 119  -0.023  0.9997
##  VR - VR Monitor Stereo         -0.065000 0.0408 119  -1.593  0.2528
##  VR Monitor - VR Monitor Stereo -0.064103 0.0430 119  -1.489  0.2997
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
# determine the mean accuracy for question types 1, 2 and 3
accplot2 <- all_data_no_outliers %>%
   group_by(ParticipantId, TrialType) %>%
   summarize(overallmean = mean(Correct_Avg)) %>%
   group_by(TrialType) %>%
   summarize(overall_trialtype_mean = mean(overallmean),
             se = std.error(overallmean),
             n = n(),
             CI = qt(0.975,df=n-1)*se)
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

```
 ggplot(accplot2, aes(TrialType,
                          overall_trialtype_mean,
                          group = 1,
                          ymin = overall_trialtype_mean - CI,
                          ymax = overall_trialtype_mean + CI)) +
   theme_classic() +
   geom_point(size = 4) +
   geom_errorbar(width = .15, linewidth = 0.85) +
   geom_line(linewidth = 0.85) +
   labs(y = "Accuracy", title = "Accuracy")
```




```
kable(accplot2)
```

TrialType	overall_trialtype_mean	se	n	CI
Type1	0.9416667	0.0166667	120	0.0330017
Type2	0.9579832	0.0184693	119	0.0365742
Type3	0.9528689	0.0165462	122	0.0327576